V. EDA – DATA VISUALIZATION

AIM:

To explore and understand the underlying patterns, distributions, and relationships within the dataset through visual representations, which aids in uncovering insights, detecting anomalies, and guiding further analysis.

PROCEDURE:

- Import the required Python libraries, such as Pandas, Matplotlib, and Seaborn.
- Load the dataset into a Pandas DataFrame.
- Inspect the dataset to understand its features and data types.
- Create visualizations such as histograms, bar charts, scatter plots, and heatmaps to identify patterns and relationships.
- Analyze the visualizations to detect anomalies, understand data distributions, and draw useful insights.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files

# Upload IPL dataset
uploaded = files.upload()

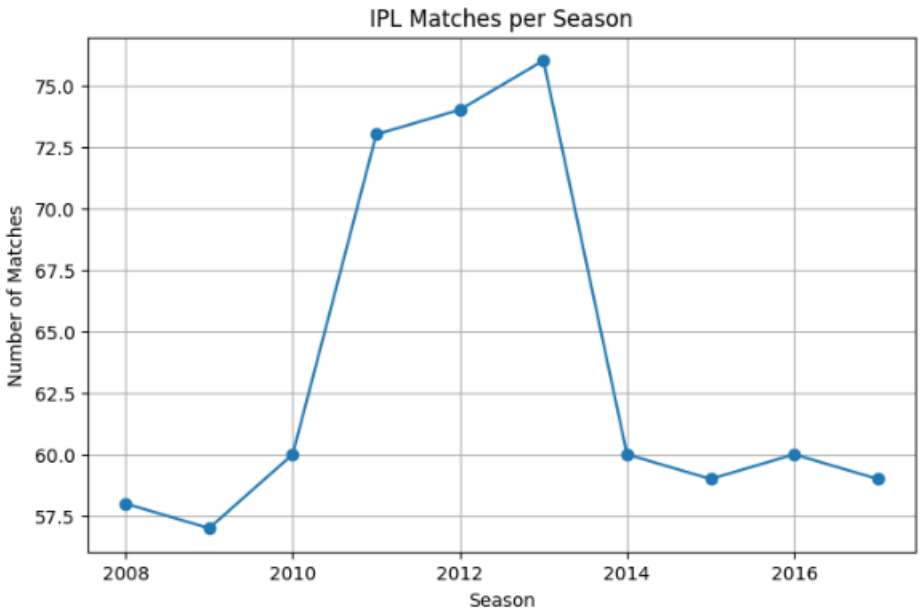
# Read dataset
df = pd.read_csv(list(uploaded.keys())[0])

print("Dataset loaded successfully!")
print(df.head())
```

```
Saving matches.csv to matches.csv
Dataset loaded successfully!
   id  season       city      date                       team1  \
0   1    2017  Hyderabad  5/4/2017          Sunrisers Hyderabad
1   2    2017       Pune  6/4/2017               Mumbai Indians
2   3    2017     Rajkot  7/4/2017                Gujarat Lions
3   4    2017     Indore  8/4/2017       Rising Pune Supergiant
4   5    2017  Bangalore  8/4/2017  Royal Challengers Bangalore
```

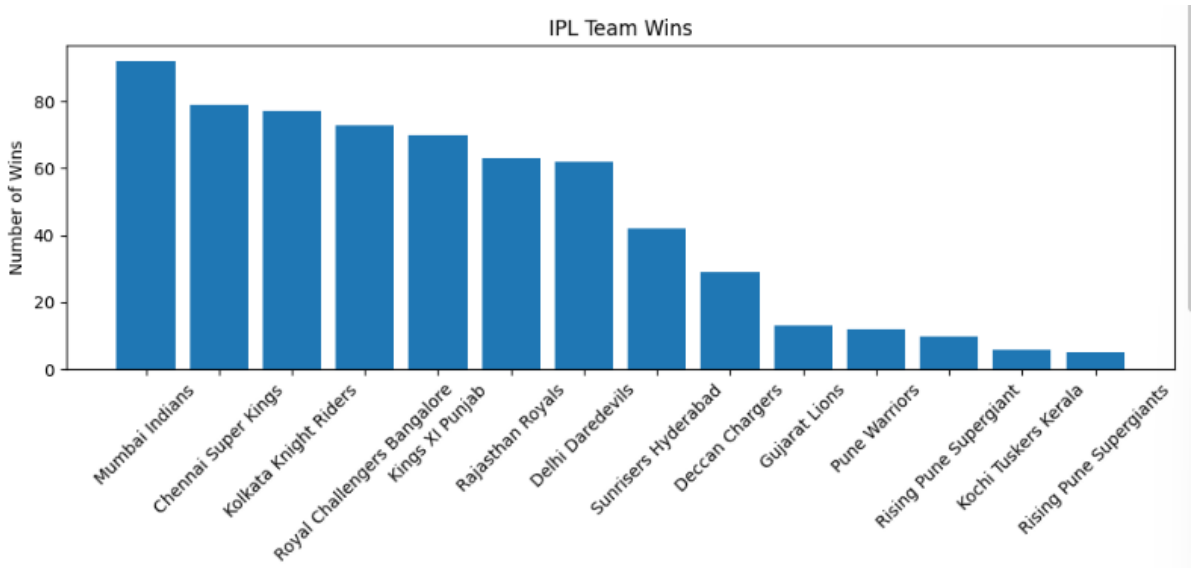
LINE CHART:

```
matches = df['season'].value_counts().sort_index()
plt.figure(figsize=(8,5))
plt.plot(matches.index, matches.values, marker='o')
plt.title("IPL Matches per Season")
plt.xlabel("Season")
plt.ylabel("Number of Matches")
plt.grid()
plt.show()
```



BAR CHART:

```
wins = df['winner'].value_counts()
plt.figure(figsize=(10,5))
plt.bar(wins.index, wins.values)
plt.title("IPL Team Wins")
plt.xlabel("Team")
plt.ylabel("Number of Wins")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



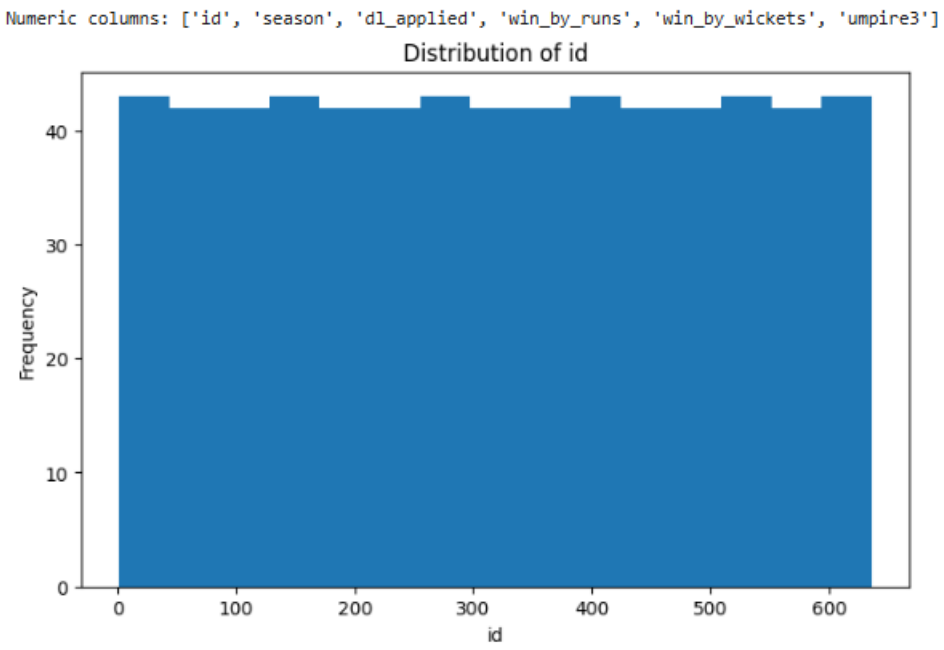
HISTOGRAM:

```
# Select a numeric column automatically
numeric_columns = df.select_dtypes(include='number').columns

print("\nNumeric columns:", numeric_columns.tolist())

# Use the first numeric column
column = numeric_columns[0]

plt.figure(figsize=(8,5))
plt.hist(df[column].dropna(), bins=15)
plt.title("Distribution of " + column)
plt.xlabel(column)
plt.ylabel("Frequency")
plt.show()
```



RESULT:

Thus, the given program was written and executed successfully.